POLY PMNA · DIPLOMA CURRICULUM REVISION 2026

Course 6012 | Semester VI | Theory | 4 Credits

Environmental Engineering

A complete syllabus-aligned handbook with clear explanations, practical or exam guidance, Malayalam-support notes, diagrams, activities and revision tools.

Programmes	**Civil Engineering**
Contact hours	**60**
Teaching scheme	**3:1:0:0**
Credits	**4**
CIE / ESE	**Theory CIE 40 / Theory ESE 60**
Self-learning	**5.5 h/week**

CURRICULUM INTEGRITY

Official Source Declaration and Verified Inconsistencies

The attached Revision 2026 index identifies Course 6012 as Environmental Engineering in Semester VI for Civil Engineering. The official SITTTR detailed course page was checked at the indexed link and returned “Requested file not found.” With the user's approved public-source approach, this handbook is transparently based on NPTEL IIT Roorkee’s Basic Environmental Engineering and Pollution Abatement course and the MSBTE polytechnic syllabus. It does not claim to reproduce official REV2026 detailed-syllabus wording.

Source treatment: Teaching scheme, credits and assessment values are provisional placeholders aligned to neighbouring handbook format. Environmental impact assessments, treatment plant designs, pollution monitoring reports and regulatory compliance decisions must be based on approved environmental standards, authorised laboratory testing, current legislation and competent professional review.

Verified source note: Verified sources support ecology, environment/biodiversity, pollution types, environmental standards, sampling, environmental laws, air pollution control, water/wastewater treatment, sludge management, industrial pollution, solid/hazardous waste and environmental legislation. The four-module sequence is a learning path, not an asserted official module split.

COURSE DASHBOARD

What You Are Expected to Learn

60
Contact hours

4
Credits

Theory CIE 40
CIE

Theory ESE 60
ESE

Course introduction

Environmental Engineering studies the protection of human health and the environment through the management of water, air, soil and waste resources. It develops the ability to design water supply and sewage treatment systems, control air and noise pollution, manage solid and hazardous waste, and apply environmental legislation and sanitation principles while respecting the technical and regulatory boundaries of environmental protection.

> **Malayalam support:** □ handbook □□□□□□□□□□□□ syllabus topic □□□□□ □□□□□□□□□□; □□□□□□□□□ principle, diagram/procedure, application, common mistake □□□□□□ □□□□□□□□□ □□□□□□□□□.

Prerequisite foundation

Basic chemistry, biology, hydraulics, environmental awareness, engineering mathematics and measurement fundamentals.

Use prerequisites only as a revision bridge. The current course outcomes and detailed syllabus define the required performance.

Teaching and assessment scheme

L	T	P	S	Self-learning/week	Contact hours	Credits	CIE	ESE
3	1	0	0	5.5 h/week	60	4	Theory CIE 40	Theory ESE 60

STUDY METHOD

How to Use This Handbook

1 · Understand

Read terms, conditions and purpose; make a short definition in your own words.

2 · Visualise

Follow the diagram, circuit, flow or procedure and label it accurately.

3 · Apply

Use the method block, calculation or practical checklist without skipping units or safety checks.

4 · Revise

Use questions, quick cards and common-mistake notes before examination.

OFFICIAL STATEMENTS

Objectives and Course Outcomes

Official course objectives

1. Explain environmental fundamentals, ecology, pollution types and the regulatory framework for environmental protection.
2. Explain water supply and wastewater engineering, including sampling, characterisation and treatment processes.
3. Explain air and noise pollution causes, effects and control technologies.
4. Explain solid and hazardous waste management, environmental sanitation and the role of environmental impact assessment.

Student-friendly meaning

You should be able to recognise the relevant system or material, explain its principle, communicate through a correct diagram/table where needed and follow a defensible solution or practical method.

> **Malayalam support:** CO □□□□□□ exam-□ □□□□□ □□□□□□□□ □□□□□□□ □□□□□ □□□□□□□□□□. Define, explain, apply □□□□□ action words □□□□□□□□□□□□□□.

Official Course Outcomes		
CO	**Official outcome**	**Bloom level**
CO1	Explain ecology, environment, pollution types and the regulatory/legal framework for environmental protection.	Understand
CO2	Explain water and wastewater characterisation and the primary, secondary and tertiary treatment processes.	Understand
CO3	Explain air and noise pollution sources, impacts and the principles of pollution control equipment.	Understand
CO4	Explain solid/hazardous waste management, environmental sanitation and environmental impact assessment principles.	Understand

Official CO-PO articulation matrix

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7–PO11
CO1	3	2	2	2	2	–	–
CO2	3	2	2	2	2	–	–
CO3	3	2	2	2	2	–	–
CO4	3	2	2	2	2	–	–

SYLLABUS COVERAGE

Curriculum Coverage Map

All official major topics mapped to handbook chapters					
Module	**Title**	**Official major topics**	**Hours**	**CO**	**Handbook section**
1	Ecology, Pollution and Environmental Legislation	Introduction to ecology and ecosystems; biodiversity and ecosystem services; pollution types and sources; transmission of pollutants; environmental standards and the legal/regulatory framework.	Approx. 14 hours	CO1	Module 1
2	Water and Wastewater Engineering and Treatment	Water and wastewater sampling and characterisation; drinking water treatment schemes; domestic and industrial wastewater treatment; primary, secondary and tertiary treatment processes; sludge management.	Approx. 16 hours	CO2	Module 2
3	Air and Noise Pollution and Control	Air pollution sources and impacts; air quality survey and monitoring; air pollution control equipment (cyclones, filters, scrubbers); noise pollution sources, measurement and control strategies.	Approx. 16 hours	CO3	Module 3
4	Solid Waste Management and Environmental Assessment	Solid waste types and management (collection, transfer, disposal); hazardous waste management; environmental sanitation; role of Environmental Impact Assessment (EIA) in project planning.	Approx. 14 hours	CO4	Module 4

LEARNING ROADMAP

How the Modules Connect

Ecology, Pollution and Environmental Legislation

Introduction to ecology and ecosystems; biodiversity and ecosystem services; pollution types and sources; transmission of pollutants; environmental standards and the legal/regulatory framework.

CO1 · Approx. 14 hours

Water and Wastewater Engineering and Treatment

Water and wastewater sampling and characterisation; drinking water treatment schemes; domestic and industrial wastewater treatment; primary, secondary and tertiary treatment processes; sludge management.

CO2 · Approx. 16 hours

Air and Noise Pollution and Control

Air pollution sources and impacts; air quality survey and monitoring; air pollution control equipment (cyclones, filters, scrubbers); noise pollution sources, measurement and control strategies.

CO3 · Approx. 16 hours

Solid Waste Management and Environmental Assessment

Solid waste types and management (collection, transfer, disposal); hazardous waste management; environmental sanitation; role of Environmental Impact Assessment (EIA) in project planning.

CO4 · Approx. 14 hours

MODULE 1

Ecology, Pollution and Environmental Legislation

Introduction to ecology and ecosystems; biodiversity and ecosystem services; pollution types and sources; transmission of pollutants; environmental standards and the legal/regulatory framework.

Approx. 14 hours **CO1** **Understand · Apply**

Learning goals

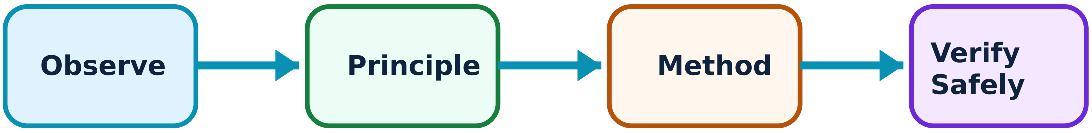
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Ecology, Pollution and Environmental Legislation: identify the system, state its principle, follow the correct method and verify the result safely.

1. Ecology, environment and biodiversity−

Ecology studies the interaction between living organisms and their environment. Biodiversity provides essential ecosystem services (water purification, climate regulation). Understanding these systems is fundamental to identifying and mitigating human-induced environmental impacts.

Study and application method

Sketch a simple ecosystem showing energy flow and nutrient cycling; describe three ecosystem services and the risks they face from human activity.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Sketch a simple ecosystem showing energy flow and nutrient cycling; describe three ecosystem services and the risks they face from human activity.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Ecosystems are complex and non-linear. Avoid assuming that one environmental intervention will not have unintended consequences elsewhere.

2. Pollution types and environmental laws−

Pollution (water, air, soil, noise) arises from diverse sources and transmits through environmental media. Environmental laws and standards (like NAAQI for air) provide the regulatory framework for monitoring, compliance and enforcement to protect public health.

Study and application method

List four major environmental acts in India and identify the primary standard for air quality in a residential area.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

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Common mistake / precaution: Environmental standards and laws are subject to frequent updates. Always verify current local and national regulations before assessing compliance.

Module 1 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

MODULE 2

Water and Wastewater Engineering and Treatment

Water and wastewater sampling and characterisation; drinking water treatment schemes; domestic and industrial wastewater treatment; primary, secondary and tertiary treatment processes; sludge management.

Approx. 16 hours | CO2 | Understand · Apply

Learning goals

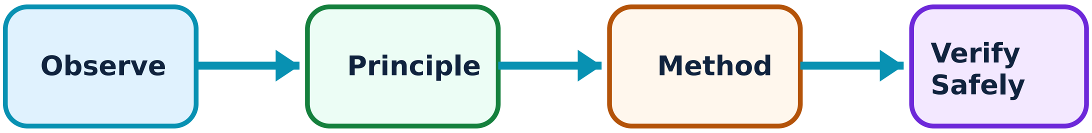
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Water and Wastewater Engineering and Treatment: identify the system, state its principle, follow the correct method and verify the result safely.

1. Water and wastewater characterisation−

Water and wastewater are characterised by physical (turbidity, colour), chemical (pH, BOD, COD, dissolved solids) and biological (coliforms) parameters. Sampling must be representative to accurately assess treatment needs and effluent quality.

Study and application method

Calculate the BOD of a wastewater sample conceptually and explain the difference between BOD and COD in assessing organic load.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Calculate the BOD of a wastewater sample conceptually and explain the difference between BOD and COD in assessing organic load.

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Common mistake / precaution: Inaccurate sampling or analysis can lead to treatment failure or environmental contamination. Use only authorised laboratory methods and representative samples.

2. Treatment processes and sludge management−

Treatment involves primary (physical), secondary (biological - e.g., activated sludge, trickling filters) and tertiary (advanced - e.g., disinfection, nutrient removal) processes. Sludge produced must be stabilised, dewatered and disposed of safely.

Study and application method

Draw a flow diagram for a domestic sewage treatment plant and label the primary, secondary and tertiary stages; describe two methods for sludge disposal.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Draw a flow diagram for a domestic sewage treatment plant and label the primary, secondary and tertiary stages; describe two methods for sludge disposal.

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Common mistake / precaution: Treatment plant operation requires skilled supervision and regular monitoring. Effluent must meet authorised standards before discharge into the environment.

Module 2 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

MODULE 3

Air and Noise Pollution and Control

Air pollution sources and impacts; air quality survey and monitoring; air pollution control equipment (cyclones, filters, scrubbers); noise pollution sources, measurement and control strategies.

Approx. 16 hours **CO3** **Understand · Apply**

Learning goals

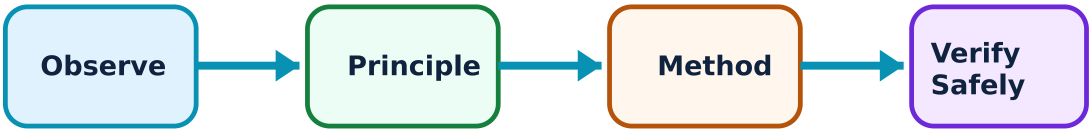
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Air and Noise Pollution and Control: identify the system, state its principle, follow the correct method and verify the result safely.

1. Air pollution monitoring and control–

Air pollutants (particulates, gases) are monitored through ambient air quality surveys. Control equipment like cyclones, fabric filters, electrostatic precipitators and scrubbers are selected based on pollutant characteristics and removal efficiency.

Study and application method

Compare the working principles and suitability of a cyclone separator and an electrostatic precipitator for particulate removal.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Compare the working principles and suitability of a cyclone separator and an electrostatic precipitator for particulate removal.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Air pollution control equipment must be properly sized and maintained. Incorrect operation can lead to excessive emissions and health risks for the surrounding community.

2. Noise pollution and management−

Noise pollution arises from transport, industry and social activities. It is measured in decibels (dB) and managed through source control, path modification (barriers) and receiver protection, guided by noise level standards for different zones.

Study and application method

Calculate the equivalent noise level conceptually and list three strategies for reducing noise from an industrial facility.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Calculate the equivalent noise level conceptually and list three strategies for reducing noise from an industrial facility.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Noise impacts are cumulative and can lead to significant health issues. Use only calibrated sound level meters for authoritative monitoring.

Module 3 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

MODULE 4

Solid Waste Management and Environmental Assessment

Solid waste types and management (collection, transfer, disposal); hazardous waste management; environmental sanitation; role of Environmental Impact Assessment (EIA) in project planning.

Approx. 14 hours | **CO4** | **Understand · Apply**

Learning goals

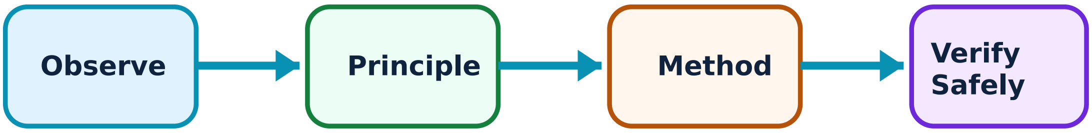
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Solid Waste Management and Environmental Assessment: identify the system, state its principle, follow the correct method and verify the result safely.

1. Solid and hazardous waste management−

Solid waste management follows a hierarchy of reduce, reuse, recycle and safe disposal (landfilling, incineration). Hazardous waste requires specific handling, treatment and disposal (TSDF) to prevent severe environmental and health impacts.

Study and application method

Outline a solid waste management plan for a small town, including segregation at source and a safe disposal method; identify three categories of hazardous waste.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Outline a solid waste management plan for a small town, including segregation at source and a safe disposal method; identify three categories of hazardous waste.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Improper waste disposal leads to groundwater pollution and disease. Do not handle or dispose of hazardous waste without appropriate training and authorisation.

2. Environmental sanitation and EIA−

Environmental sanitation (rural/urban) focuses on safe excreta disposal and hygiene. Environmental Impact Assessment (EIA) is a planning tool that predicts the environmental consequences of a proposed project and identifies mitigation measures.

Study and application method

Describe two methods of low-cost rural sanitation and outline the major steps in an Environmental Impact Assessment for a new highway project.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Describe two methods of low-cost rural sanitation and outline the major steps in an Environmental Impact Assessment for a new highway project.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: EIA is a formal legal requirement for many projects. It must be conducted by accredited professionals and include public consultation as per regulations.

Module 4 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

FORMULA AND PROCEDURE BANK

Rapid Recall Bank

Recall 1

State symbols and SI units before substitution.

Recall 2

Draw the circuit, system boundary, vector or process flow before reasoning.

Recall 3

For laboratory work: aim → apparatus → diagram → procedure → observation → calculation → result → precautions.

Recall 4

For a graph: label axes, units, scale, operating region and conclusion.

Recall 5

For a comparison: use construction, working, result, advantage and limitation.

Recall 6

For an extended answer: definition/principle, labelled diagram, working steps, application and a caution/limitation.

DIAGRAM LIBRARY

Diagram Library for Rapid Revision

Module 1: Ecology, Pollution and Environmental Legislation

Use the concept flow to revise observation, principle, method and verification.

Module 2: Water and Wastewater Engineering and Treatment

Use the concept flow to revise observation, principle, method and verification.

Module 3: Air and Noise Pollution and Control

Use the concept flow to revise observation, principle, method and verification.

Module 4: Solid Waste Management and Environmental Assessment

Use the concept flow to revise observation, principle, method and verification.

SELF-LEARNING

Official Self-Learning and Student Activities

Structured activity

Sketch an ecosystem energy-flow diagram and identify two human-induced risks.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

Draw a flow chart for a domestic water treatment plant and a sewage treatment plant.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

Compare three air pollution control devices based on their removal mechanism and application.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

Prepare a solid waste segregation and disposal plan for a hypothetical residential colony.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

Outline the steps for conducting an Environmental Impact Assessment for a local infrastructure project.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Activity discipline: The supplied PDF assigns the listed self-learning hours. This handbook preserves the activity direction; the teacher’s approved schedule and safety instructions govern execution.

EXAMINATION READINESS

Assessment Methodology and Examination Pattern

Official assessment structure

Component	Official mark / conversion	Student evidence
Attendance	5 marks	End-of-semester attendance component.
Self-learning activities	15 marks	Module-linked activities.
Series examinations	20 + 20 converted	Two 50-mark series examinations.
CIE total	Theory CIE 40	Continuous internal evaluation.
Written ESE	Theory ESE 60	2.5 hours; official Part A / Part B / Part C pattern.

Series-test pattern

The supplied theory pattern uses Part A (1-mark), Part B (3-mark with choice) and Part C (7-mark) distribution across the stated modules. Practical courses follow the supplied practical-test scheme.

Examination evidence

Show a complete method, correct units/conditions, clear diagrams or tables, a result/conclusion and safety awareness for any apparatus or process involved.

EXAM PRACTICE

Module-wise Questions with Answers

Module 1 — Ecology, Pollution and Environmental Legislation

Explain Ecology, environment and biodiversity and state one correct method, application or precaution.−

Ecology studies the interaction between living organisms and their environment. Biodiversity provides essential ecosystem services (water purification, climate regulation). Understanding these systems is fundamental to identifying and mitigating human-induced environmental impacts.

Answer framework: Sketch a simple ecosystem showing energy flow and nutrient cycling; describe three ecosystem services and the risks they face from human activity.

Important point: Ecosystems are complex and non-linear. Avoid assuming that one environmental intervention will not have unintended consequences elsewhere.

Explain Pollution types and environmental laws and state one correct method, application or precaution.−

Pollution (water, air, soil, noise) arises from diverse sources and transmits through environmental media. Environmental laws and standards (like NAAQI for air) provide the regulatory framework for monitoring, compliance and enforcement to protect public health.

Answer framework: List four major environmental acts in India and identify the primary standard for air quality in a residential area.

Important point: Environmental standards and laws are subject to frequent updates. Always verify current local and national regulations before assessing compliance.

Module 2 — Water and Wastewater Engineering and Treatment

Explain Water and wastewater characterisation and state one correct method, application or precaution.−

Water and wastewater are characterised by physical (turbidity, colour), chemical (pH, BOD, COD, dissolved solids) and biological (coliforms) parameters. Sampling must be representative to accurately assess treatment needs and effluent quality.

Answer framework: Calculate the BOD of a wastewater sample conceptually and explain the difference between BOD and COD in assessing organic load.

Important point: Inaccurate sampling or analysis can lead to treatment failure or environmental contamination. Use only authorised laboratory methods and representative samples.

Explain Treatment processes and sludge management and state one correct method, application or precaution.−

Treatment involves primary (physical), secondary (biological - e.g., activated sludge, trickling filters) and tertiary (advanced - e.g., disinfection, nutrient removal) processes. Sludge produced must be stabilised, dewatered and disposed of safely.

Answer framework: Draw a flow diagram for a domestic sewage treatment plant and label the primary, secondary and tertiary stages; describe two methods for sludge disposal.

Important point: Treatment plant operation requires skilled supervision and regular monitoring. Effluent must meet authorised standards before discharge into the environment.

Module 3 — Air and Noise Pollution and Control

Explain Air pollution monitoring and control and state one correct method, application or precaution.−

Air pollutants (particulates, gases) are monitored through ambient air quality surveys. Control equipment like cyclones, fabric filters, electrostatic precipitators and scrubbers are selected based on pollutant characteristics and removal efficiency.

Answer framework: Compare the working principles and suitability of a cyclone separator and an electrostatic precipitator for particulate removal.

Important point: Air pollution control equipment must be properly sized and maintained. Incorrect operation can lead to excessive emissions and health risks for the surrounding community.

Explain Noise pollution and management and state one correct method, application or precaution.−

Noise pollution arises from transport, industry and social activities. It is measured in decibels (dB) and managed through source control, path modification (barriers) and receiver protection, guided by noise level standards for different zones.

Answer framework: Calculate the equivalent noise level conceptually and list three strategies for reducing noise from an industrial facility.

Important point: Noise impacts are cumulative and can lead to significant health issues. Use only calibrated sound level meters for authoritative monitoring.

Module 4 — Solid Waste Management and Environmental Assessment

Explain Solid and hazardous waste management and state one correct method, application or precaution.−

Solid waste management follows a hierarchy of reduce, reuse, recycle and safe disposal (landfilling, incineration). Hazardous waste requires specific handling, treatment and disposal (TSDF) to prevent severe environmental and health impacts.

Answer framework: Outline a solid waste management plan for a small town, including segregation at source and a safe disposal method; identify three categories of hazardous waste.

Important point: Improper waste disposal leads to groundwater pollution and disease. Do not handle or dispose of hazardous waste without appropriate training and authorisation.

Explain Environmental sanitation and EIA and state one correct method, application or precaution.–

Environmental sanitation (rural/urban) focuses on safe excreta disposal and hygiene. Environmental Impact Assessment (EIA) is a planning tool that predicts the environmental consequences of a proposed project and identifies mitigation measures.

Answer framework: Describe two methods of low-cost rural sanitation and outline the major steps in an Environmental Impact Assessment for a new highway project.

Important point: EIA is a formal legal requirement for many projects. It must be conducted by accredited professionals and include public consultation as per regulations.

PRACTICE ONLY

Practice Model Paper

Important: This practice paper is based on the official pattern in the supplied syllabus. It is **not** an official SITTTR model question paper.

Part A — short response

1. State one key definition from Module 1: Ecology, Pollution and Environmental Legislation.
2. Identify one diagram, observation, formula or safety condition relevant to Module 1.
3. State one key definition from Module 2: Water and Wastewater Engineering and Treatment.
4. Identify one diagram, observation, formula or safety condition relevant to Module 2.
5. State one key definition from Module 3: Air and Noise Pollution and Control.
6. Identify one diagram, observation, formula or safety condition relevant to Module 3.
7. State one key definition from Module 4: Solid Waste Management and Environmental Assessment.
8. Identify one diagram, observation, formula or safety condition relevant to Module 4.

Part B — structured explanation

1. Explain a selected procedure/principle from Module 1 with a labelled diagram, table or method.
2. Explain a selected procedure/principle from Module 2 with a labelled diagram, table or method.
3. Explain a selected procedure/principle from Module 3 with a labelled diagram, table or method.
4. Explain a selected procedure/principle from Module 4 with a labelled diagram, table or method.

Part C — extended response / practical task

1. Develop a complete answer or practical plan for: Introduction to ecology and ecosystems; biodiversity and ecosystem services; pollution types and sources; transmission of pollutants; environmental standards and the legal/regulatory framework..
2. Develop a complete answer or practical plan for: Water and wastewater sampling and characterisation; drinking water treatment schemes; domestic and industrial wastewater treatment; primary, secondary and tertiary treatment processes; sludge management..
3. Develop a complete answer or practical plan for: Air pollution sources and impacts; air quality survey and monitoring; air pollution control equipment (cyclones, filters, scrubbers); noise pollution sources, measurement and control strategies..
4. Develop a complete answer or practical plan for: Solid waste types and management (collection, transfer, disposal); hazardous waste management; environmental sanitation; role of Environmental Impact Assessment (EIA) in project planning..

LAST-MILE REVISION

Quick Revision

Module 1: Ecology, Pollution and Environmental Legislation

Introduction to ecology and ecosystems; biodiversity and ecosystem services; pollution types and sources; transmission of pollutants; environmental standards and the legal/regulatory framework.

- Match each topic to CO1.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Module 2: Water and Wastewater Engineering and Treatment

Water and wastewater sampling and characterisation; drinking water treatment schemes; domestic and industrial wastewater treatment; primary, secondary and tertiary treatment processes; sludge management.

- Match each topic to CO2.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Module 3: Air and Noise Pollution and Control

Air pollution sources and impacts; air quality survey and monitoring; air pollution control equipment (cyclones, filters, scrubbers); noise pollution sources, measurement and control strategies.

- Match each topic to CO3.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Module 4: Solid Waste Management and Environmental Assessment

Solid waste types and management (collection, transfer, disposal); hazardous waste management; environmental sanitation; role of Environmental Impact Assessment (EIA) in project planning.

- Match each topic to CO4.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Common mistakes: Writing an unlabelled diagram; ignoring units, polarity or conditions; treating a practical result as valid without observations; confusing a definition with an application; and omitting a safety precaution where the topic uses apparatus, current, heat, chemicals or tools.

Exam checklist: Read the command word; answer every part; draw clean labelled diagrams; use a clear calculation/procedure; state result with units or observation; and reserve two minutes for checking.

VOCABULARY

Glossary

Important terms from the syllabus	
Term / topic	**One-line meaning**
Ecology, environment and biodiversity	Ecology studies the interaction between living organisms and their environment.
Pollution types and environmental laws	Pollution (water, air, soil, noise) arises from diverse sources and transmits through environmental media.
Water and wastewater characterisation	Water and wastewater are characterised by physical (turbidity, colour), chemical (pH, BOD, COD, dissolved solids) and biological (coliforms) parameters.
Treatment processes and sludge management	Treatment involves primary (physical), secondary (biological - e.
Air pollution monitoring and control	Air pollutants (particulates, gases) are monitored through ambient air quality surveys.
Noise pollution and management	Noise pollution arises from transport, industry and social activities.
Solid and hazardous waste management	Solid waste management follows a hierarchy of reduce, reuse, recycle and safe disposal (landfilling, incineration).
Environmental sanitation and EIA	Environmental sanitation (rural/urban) focuses on safe excreta disposal and hygiene.

LEARNING RESOURCES

References

Recommended environmental-engineering references

1. S. K. Garg, Environmental Engineering (Vol. I & II).
2. Peavy, Rowe, and Tchobanoglous, Environmental Engineering.
3. Metcalf & Eddy, Wastewater Engineering: Treatment and Resource Recovery.
4. M. N. Rao and H. V. N. Rao, Air Pollution.

Approved public environmental-engineering sources

- https://nptel.ac.in/courses/103107215
- https://onlinecourses.nptel.ac.in/noc26_ch73/preview

These sources provide the verified study basis while official Course 6012 detail is unavailable; they do not replace official environmental standards, authorised compliance reports, project-specific EIA or legal regulatory requirements.